GATE DA 2027

Prediction & Pattern Report

What three years of GATE DA papers say about the fourth

Companion to **Predicted Mock Paper 1** and **Predicted Mock Paper 2**

Organising institute: **IIT Madras** • Exam dates: **6, 7, 13, 14, 20, 21 February 2027**

A word on what this document is. GATE DA has existed for three sittings only, so this is a forecast built on a short series, not a law of nature. Every claim below is labelled either as a *verified fact* (from official announcements or published paper analyses) or as *my inference*. Treat the topic forecast as a way to order your revision, never as a licence to skip syllabus.

Prepared for Ashutosh • August 2026

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The six calls, up front

1. **The DA syllabus does not change for 2027.** IIT Madras revised syllabi across GATE 2027 papers, but the DA syllabus is identical to 2026. Everything you have already prepared still counts. *(Verified.)*

2. **Mathematics — Probability & Statistics plus Linear Algebra plus Calculus & Optimization — will carry 32 to 38 marks.** It has never gone below 31 in three papers. This is the single largest block, larger than ML and AI combined. *(My inference from the three papers.)*

3. **MSQs keep rising, NATs keep falling.** 10 → 12 → 14 MSQs and 25 → 21 → 18 NATs across 2024–2026. Expect roughly 33 MCQ, 14 MSQ, 18 NAT. This matters: MSQs carry no negative marking but give no partial credit either. *(Verified counts, my extrapolation.)*

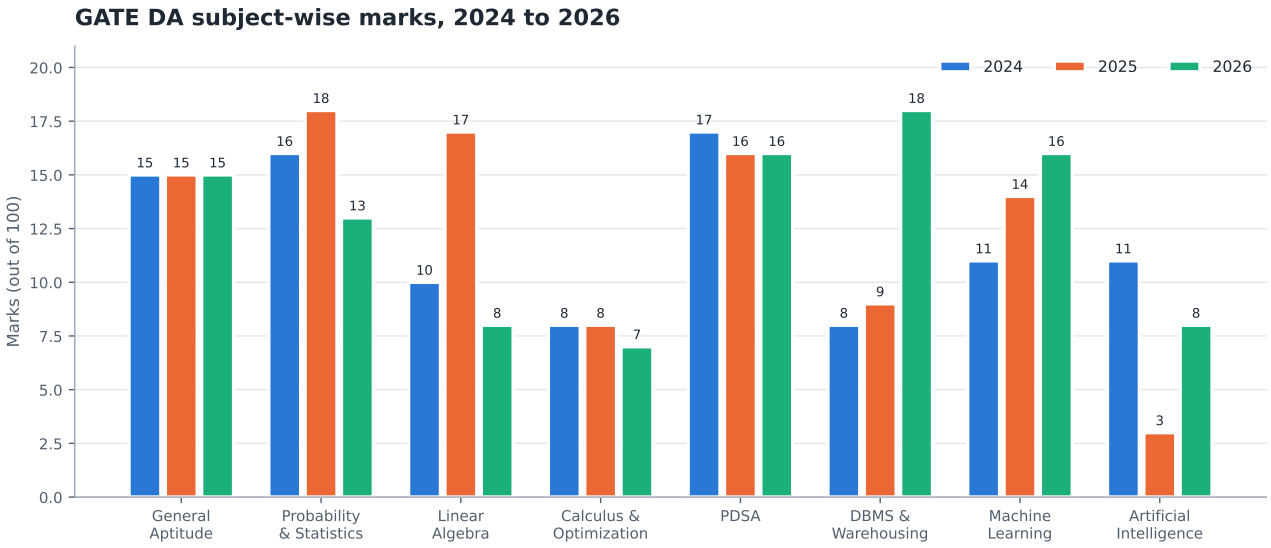
4. **One subject will spike and one will collapse — you cannot predict which.** AI went 11 → 3 → 8; DBMS went 8 → 9 → 18; Linear Algebra went 10 → 17 → 8. Volatility, not weightage, is the real story of this paper. Do not abandon a "low-weightage" subject. *(Verified.)*

5. **Bayes' theorem, eigenvalues, a Python trace, an SQL/normalization question and a metrics/overfitting question are close to guaranteed.** All five have appeared in every paper so far. *(Verified across 2024–2026.)*

6. **Expect the paper to be calculation-light and property-heavy.** All three papers have rewarded knowing a property (memorylessness, admissibility, stability of a sort, closure of an FD set) over grinding arithmetic. *(Consistent commentary across published analyses.)*

1. The three papers so far

1.1 Subject-wise marks



Subject	2024	2025	2026	Mean
General Aptitude	15	15	15	15.0
Probability & Statistics	16	18	13	15.7
Linear Algebra	10	17	8	11.7
Calculus & Optimization	8	8	7	7.7
Programming, Data Structures & Algorithms	17	16	16	16.3
Database Management & Warehousing	8	9	18	11.7
Machine Learning	11	14	16	13.7
Artificial Intelligence	11	3	8	7.3

The 2024 and 2026 columns come from independent published paper analyses and do not sum to exactly 100 — different analysts assign a few borderline questions to different subjects (a PCA question is both Linear Algebra and Machine Learning; a covariance question is both Probability and Machine Learning). Treat each cell as ±2 marks. The 2025 column sums exactly.

1.2 What the three papers agree on

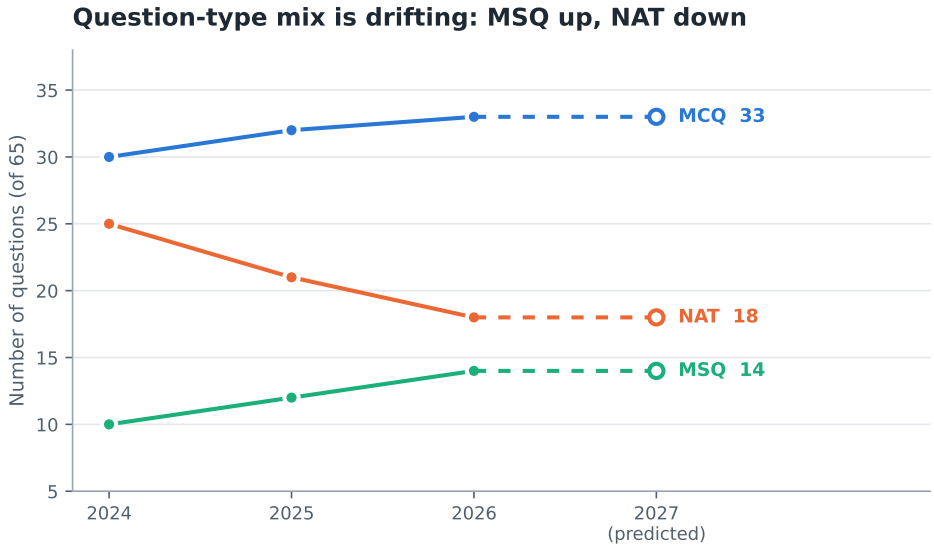
- **PDSA is the most stable subject in the paper** — 17, 16, 16. It is also the most time-expensive. If you are strong here you bank 16 marks with near-certainty.
- **Probability & Statistics is the largest maths block** in all three papers and has never dropped below 13.
- **General Aptitude is exactly 15 marks every year** — 5 one-mark and 5 two-mark questions. It is also the cheapest 15 marks in the paper and the most commonly under-prepared.
- **Calculus & Optimization is consistently small** (7–8) and consistently present. Do not over-invest, do not skip.

1.3 What they disagree on — and why that matters more

Three of the seven core subjects have swung by 8 marks or more between consecutive papers. That is a much bigger effect than the difference between the "high-weightage" and "low-weightage" subjects.

In practice this means the winning strategy for 2027 is **breadth with no holes**, not depth in the top three subjects. A candidate who is strong in ML and Probability but blank in AI lost real marks in 2024 (AI was 11) and would have been fine in 2025 (AI was 3) — that is a coin flip you do not want to be exposed to.

1.4 Question types



The drift is monotone in both directions across three years, which is why I am willing to extrapolate it. The practical consequence:

> **More MSQs is a difficulty increase in disguise.** An MSQ has no negative marking, but it also has no partial credit — you must get *every* option right. A four-option MSQ is effectively four true/false judgements that must all land. Practise reading each option as a standalone claim; that is a different skill from eliminating three wrong choices in an MCQ.

2. Who is setting the paper

2.1 The verified facts

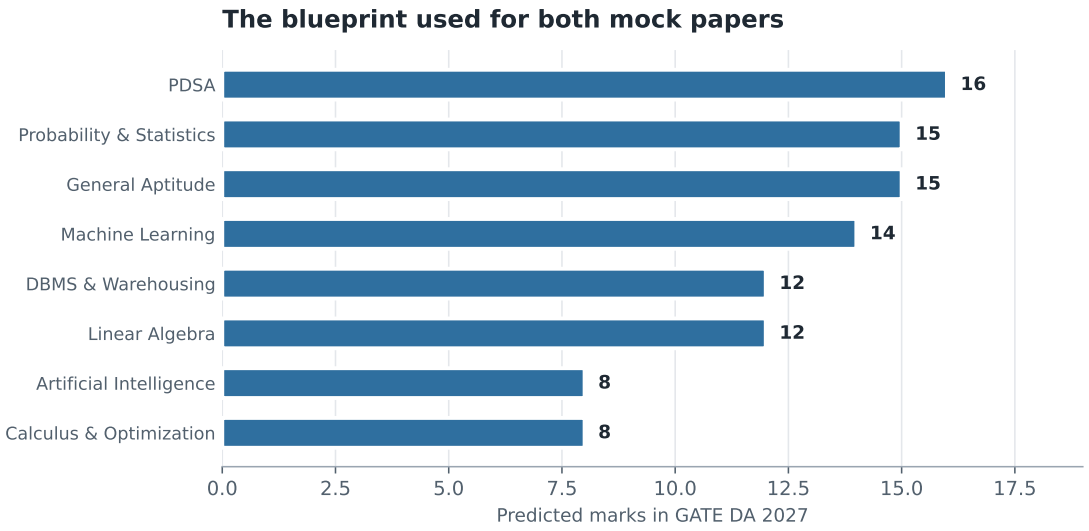
- **IIT Madras is the organising institute** for GATE 2027, the 45th edition; the official portal is `gate2027.iitm.ac.in`.
- **Exam dates: 6, 7, 13, 14, 20 and 21 February 2027**, forenoon and afternoon sessions.
- **30 test papers**, with a new **Robotics and Automation (RA)** paper added and Textile Engineering folded into the XE paper.
- **Syllabi were revised "after 5 years"** across papers — but the **DA syllabus is unchanged from 2026**. CSE saw only small deletions (secondary storage; some explicitly-named networking protocols).
- IIT Madras last organised GATE in **2019**.

2.2 What I infer from this — and how confident I am

Be sceptical of "institute X sets hard papers". GATE papers are set by subject committees drawn from across the IITs and IISc; the organising institute runs the logistics and chairs the process. The year-to-year difficulty swing in GATE DA (2024 moderate–hard, 2025 moderate–hard, 2026 moderate–slightly tough) is smaller than the noise between analysts. **I would not change a single hour of your preparation on the basis of which IIT is organising.** What follows is therefore offered as weak evidence.

- **The syllabus being frozen is the strongest signal in this whole report.** When a syllabus is untouched, setters have no mandate to introduce unfamiliar topic areas, and the previous papers become a much better guide. This raises my confidence in every forecast below.
- **IIT Madras runs the BS in Data Science and Applications and a large share of NPTEL's data-science catalogue.** The DA syllabus terminology already tracks that standard treatment closely. My weak expectation: questions phrased in standard-textbook language (Bishop / Hastie / Cormen / Korth), rather than exotic or research-flavoured framings.
- **GATE 2019 CS, the last IIT Madras edition, was widely read as ranging from easy to moderately difficult**, with a heavy load of numerical-answer questions and few pure recall questions. If any style carry-over exists, it points towards *application over memorisation* — which is also what all three DA papers have already done.
- **A fourth-edition paper tends to consolidate.** 2024 was an inaugural paper and showed it (uneven subject spread); 2025 and 2026 have moved towards a more balanced split. I expect 2027 to look more like the mean of the three than like any one of them.

3. The 2027 blueprint I am predicting



Subject	Predicted	Questions	1-mark	2-mark	Range I'd allow
General Aptitude	15	10	5	5	exactly 15
Probability & Statistics	15	10	5	5	12–18
Programming, DS & Algorithms	16	10	4	6	14–18
Machine Learning	14	9	4	5	11–17
Linear Algebra	12	8	4	4	8–17
Database Mgmt. & Warehousing	12	8	4	4	8–18
Calculus & Optimization	8	5	2	3	7–9
Artificial Intelligence	8	5	2	3	3–11
Total	**100**	**65**	**30**	**35**	

The "range I'd allow" column is the honest part of this table. Both mock papers are built on the point prediction, but you should read a subject's range as the real planning input: Linear Algebra could plausibly be worth 8 marks or 17, and you must be ready for the 17.

3.1 Predicted question-type mix

	2024	2025	2026	2027 (predicted)
MCQ (negative marking)	30	32	33	**33**
MSQ (no negative marking, no partial credit)	10	12	14	**14**
NAT (no negative marking)	25	21	18	**18**

4. Ranked topic forecast

Confidence labels below mean: **Near-certain** — appeared in all three papers, or is structurally unavoidable, so I expect at least one question with probability above roughly 80%. **Likely** — appeared in two of three, roughly 50–80%. **Possible** — appeared once or is a natural but untested corner of the syllabus, roughly 25–50%.

4.1 Probability & Statistics

Topic	Confidence	Typical framing
Bayes' theorem / conditional probability	**Near-certain**	2-mark NAT, disease-test or defect story
Named distribution properties (Normal, Exponential, Poisson, Binomial, Uniform)	**Near-certain**	1- or 2-mark, tests a property not arithmetic
Expectation, variance, covariance, correlation	**Near-certain**	often embedded in an ML question
Counting / basic probability axioms	**Likely**	single cheap 1-mark
CDF / PDF manipulation, transformations of a random variable	**Likely**	2-mark NAT
Central Limit Theorem & confidence intervals	**Likely**	new-ish; explicitly in syllabus
Hypothesis testing (z-test, t-test)	**Likely**	needs a table value; keep the critical points memorised
Chi-squared test, descriptive statistics	**Likely**	goodness-of-fit or independence
Conditional expectation $E[X \mid Y]$, law of total variance	**Likely**	2-mark, appeared in 2024
t and χ^2 *distributions* (as distributions, not tests)	**Possible**	definition / degrees of freedom

4.2 Linear Algebra

Topic	Confidence	Typical framing
Eigenvalues and eigenvectors	**Near-certain**	trace/determinant shortcuts pay off
Rank, nullity, rank–nullity theorem	**Near-certain**	1-mark or as part of a system question
Systems of linear equations, consistency	**Likely**	condition on a parameter k
Determinants and special matrices (symmetric, orthogonal, projection)	**Likely**	property MSQ
SVD, matrix decompositions, PCA link	**Likely**	bridges into Machine Learning
Vector spaces, basis, linear independence	**Possible**	conceptual MSQ

4.3 Calculus & Optimization

Topic	Confidence	Typical framing
Maxima / minima, single and multivariable	**Near-certain**	2-mark
Convexity and its consequences	**Likely**	MSQ on which functions are convex
Gradient descent as an update rule	**Likely**	one-step NAT; overlaps ML
Constrained optimization / Lagrange multipliers	**Possible**	2-mark
Limits, continuity, Taylor expansion, integration	**Possible**	cheap 1-mark when it appears

4.4 Programming, Data Structures & Algorithms

Topic	Confidence	Typical framing
Python code tracing (mutability, scoping, comprehensions)	**Near-certain**	1- or 2-mark NAT; a DA signature
Asymptotic complexity and recurrences	**Near-certain**	Master theorem or unrolling
Trees — BST, traversals, heaps	**Near-certain**	2-mark
Graph algorithms — BFS/DFS, shortest path, MST, topological order	**Near-certain**	2-mark
Sorting and searching, stability, worst cases	**Likely**	property MSQ
Hashing — collision resolution	**Likely**	2-mark trace
Recursion and dynamic programming	**Likely**	count-the-calls NAT

4.5 Database Management & Data Warehousing

Topic	Confidence	Typical framing
Functional dependencies, closures, normalization (2NF/3NF/BCNF)	**Near-certain**	2-mark, "highest normal form"
SQL query output, aggregation, `GROUP BY` / `HAVING`, sub-queries	**Near-certain**	2-mark NAT
Keys — candidate, primary, foreign	**Near-certain**	1-mark
B^+ trees — structure, order, insertion	**Likely**	spiked in 2026
Relational algebra	**Likely**	translate to/from SQL
Transactions, serializability, concurrency	**Likely**	precedence graph
Data warehousing — OLAP, star/snowflake schema, ETL	**Likely**	1-mark definitional; DA-specific
File organisation, indexing cost calculations	**Possible**	block/record arithmetic

4.6 Machine Learning

Topic	Confidence	Typical framing
Bias–variance, overfitting, regularization (L_1 vs L_2)	**Near-certain**	MSQ
Evaluation metrics — confusion matrix, precision, recall, F_1, ROC	**Near-certain**	2-mark NAT
Linear and logistic regression	**Near-certain**	closed form or one forward pass
Clustering — k-means, hierarchical	**Near-certain**	2-mark trace
PCA and dimensionality reduction	**Near-certain**	links straight to Linear Algebra
Decision trees — entropy, information gain, Gini	**Likely**	2-mark NAT
Cross-validation and model selection	**Likely**	MSQ
Neural networks — parameter counting, activations, backpropagation	**Likely**	counting NAT is the common form
SVM and kernels; Naive Bayes; k-NN	**Possible**	one of the three usually appears

4.7 Artificial Intelligence

Topic	Confidence	Typical framing
Uninformed and informed search; A^*; admissible & consistent heuristics	**Near-certain**	the most reliable AI question
Propositional and first-order logic	**Near-certain**	appeared in 2026
Adversarial search — minimax, alpha–beta pruning	**Likely**	2-mark NAT
Constraint satisfaction problems	**Possible**	
Probabilistic reasoning / Bayesian networks	**Possible**	overlaps Probability

5. How the two mock papers map to this forecast

Both papers are built to the blueprint in Section 3 exactly: 65 questions, 100 marks, 10 GA + 55 core, 30 one-mark + 35 two-mark, and the same per-subject mark split. Within each subject the questions are drawn from the **Near-certain** and **Likely** rows of Section 4, in roughly the proportion those rows are expected to appear.

Paper	Calibration
Mock Paper 1	Mirrors the observed GATE DA difficulty curve. Roughly 30% direct application of a definition, 50% true exam level, 20% multi-step. Your score here is the one that is comparable to a real GATE score.
Mock Paper 2	Deliberately pitched above the observed level — longer two-mark chains and more MSQs where every option must be judged. Treat a lower score here as expected, not as a warning sign.

5.1 Using them properly

1. Sit Paper 1 in one uninterrupted 3-hour block, with only the statistical tables to hand. Attempt it exactly as you would the real exam, including the decision to leave MCQs blank.
2. Score with GATE negative marking. Fill in the score sheet at the back — the per-subject row is the whole point of it.
3. Wait at least four days. Revise the two weakest subjects. Then sit Paper 2.
4. Compare *per-subject* scores across the two papers, not totals. A subject that drops sharply from Paper 1 to Paper 2 is one where you know the definitions but cannot yet apply them under time pressure — that is a practice problem, not a revision problem.

5.2 Time budget that the blueprint implies

Block	Questions	Marks	Suggested time
GA (Q.1–Q.10)	10	15	18 minutes
Core 1-mark (Q.11–Q.35)	25	25	40 minutes
Core 2-mark (Q.36–Q.65)	30	60	105 minutes
Review and revisit flagged questions	—	—	17 minutes
Total	**65**	**100**	**180 minutes**

6. Exam-day strategy implied by the forecast

1. **Attempt every NAT and every MSQ you have any handle on.** Neither carries negative marking. Leaving a NAT blank is strictly worse than a considered guess. This alone is worth several marks and is the most commonly wasted edge in GATE DA.
2. **On MCQs, guess only after eliminating two options.** With four options and $-1/3$, a blind guess is expectation-neutral; after eliminating one it is slightly positive; after two it is clearly positive.
3. **Do General Aptitude first, in under 18 minutes.** It is 15 marks at the lowest difficulty in the paper, and doing it first protects it from being squeezed at the end.
4. **Take the one-mark core block before the two-mark block.** 25 questions worth 25 marks that are mostly single-concept. Do not let a 2-mark graph traversal eat the time of four 1-mark definitions.
5. **Flag and leave any question that needs more than three minutes on first read.** In a paper where 30 questions are worth 2 marks each, two abandoned questions cost less than one over-run.
6. **Keep these values memorised so you never lose time to the table:** $z_{0.05} = 1.645$, $z_{0.025} = 1.960$, $z_{0.005} = 2.576$; $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$; $t_{0.025,\infty} = 1.960$; $\chi^2_{0.05,1} = 3.841$.
7. **Do not skip a subject because this report calls it low-weightage.** AI was worth 3 marks in 2025 and 11 in 2024. The forecast orders your revision; it does not licence a gap.

7. The honest limits of this forecast

- **Three data points.** Any trend fitted to three observations is fragile. The question-type drift is monotone in both directions, which is why I extrapolated it; the subject-marks series is not, which is why Section 3 gives ranges rather than points.
- **Source disagreement.** Published subject-wise breakdowns for 2024 and 2026 do not agree with each other to the mark, largely because questions that sit on a boundary (PCA, covariance, complexity of an ML algorithm) get assigned differently. I have used the consensus and flagged the uncertainty rather than picking a favourite source.
- **Organising-institute effects are weak.** I have included the IIT Madras material because you asked for it, and labelled it as weak evidence, because that is what it is.
- **A forecast cannot invent a question.** The two mock papers are written to the predicted shape and to the historically favoured topics. They are not leaked questions and no one can supply those. Their value is calibration — the shape, the time pressure, the type mix — not the specific numbers.

Sources

- IIT Madras press release announcing GATE 2027 dates, syllabus revision and the new Robotics & Automation paper — `iitm.ac.in`
- GATE 2027 official portal — `gate2027.iitm.ac.in`
- GATE 2027 CSE & DA syllabus change analysis (DA unchanged) — GO Classes; The ML Hub

- GATE DA 2026 question paper with subject-wise analysis and weightage — Piyush Wairale
- GATE DA 2026 exam analysis (question-type counts, difficulty) — Physics Wallah
- GATE DA 2025 and GATE DA 2024 paper analyses (subject-wise marks, MCQ/MSQ/NAT counts) — GeeksforGeeks